

IPsec VPN Executive Assessment

Audited Target: tmpil7vway0.pcap | Captured Packets: 2

Audit Date: 29 Sep 2026, 23:31 UTC • Regulatory Baseline: NIST SP 800-77 / RFC 7296

SECURITY RISK SCORE 35/100 MEDIUM Rating	AI TRAFFIC CLASS VPN-FT XGBoost Inference	CLASSIFIER CONFIDENCE 60.1% Softmax Probability	CIPHER SUITE 3DES unknown
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3x3 Threat Assessment Matrix

	LOW IMPACT	MED IMPACT	HIGH IMPACT
HIGH LIKELIHOOD	MEDIUM	HIGH	CRITICAL
MED LIKELIHOOD	LOW	MEDIUM	HIGH
LOW LIKELIHOOD	LOW	LOW	MEDIUM

Cryptographic & Protocol Compliance Parameters

PARAMETER	OBSERVED VALUE IN CAPTURE	COMPLIANCE BASELINE (NIST)
IPsec Protocol / IKE Version	IKEv2 (IKE_SA_INIT)	IKEv2 (RFC 7296)
Encryption Cipher & Key Size	3DES (256-bit)	AES-256-GCM / AES-256-CBC
Integrity & Hash Algorithm	unknown	AEAD / HMAC-SHA2-256
Diffie-Hellman Key Exchange	Group 19	Group 14, 19, 20, 21, 28 (≥ 2048-bit)
Security Association (SPIs)	Init: 1122334455667788 Resp: 0000000000000000	Valid 64-bit Non-Zero SPI Pair

Security Assessment & Actionable Recommendations

[IPSEC-ENC-001] Weak or forbidden encryption algorithm detected [HIGH] Observed: 3DES Expected: Approved encryption (AES-256-GCM, AES-128-GCM, AES-256-CBC, ENCR_AES_GCM_16, ENCR_AES_CBC) Remediation: Reconfigure IPsec proposals to use strong encryption algorithms such as AES-256-GCM.
[IPSEC-META-003] Static SPI Header Linkability / Session Tracking Risk [LOW] Observed: Observable ESP SPI(s): 0x00001000 Expected: Dynamic SA Rekeying & SPI Rotation Remediation: Ensure frequent Security Association rekeying to prevent passive netflow session correlation.